

CSC1005/CSC1022 Database Lab

Lab 7: SQL Practice

Goal: Practice miscellaneous SQL queries and try out new DDL clauses.

You have finished the lab when you have finished all queries.

Section 1. Basketball (Make default database)

1. Return the id of the players who in 2000 played more than 1000 minutes and in 2001 played for the Miami Heat and got more than 250 assists.

Solution:

```
select pt1.playerID, pt1.points from players_teams pt1, teams t,
players_teams pt2
where pt1.minutes > 1000 and
pt1.year = 2000 and pt1.playerID = pt2.playerID
and pt2.assists > 250 and pt2.year = 2001
and t.tmID = pt2.tmID and t.year=pt2.year
and t.name = 'Miami Heat';
```

playerID	minutes	rebounds
jonesed02	2282	378
stricro02	1020	232

2. Return the id, the name, the height, and the number of points scored by the player having height more than the average height of all the players and having scored a number of points in 2005 higher than the max number of points scored by anyone in the year 1970.

Solution:

```
select pt.playerID, p.firstName, p.lastName, p.height, pt.points from
players_teams pt, players p
where pt.year = 2005 and pt.playerID = p.playerID
and pt.points >
(select max(points) from players_teams where year = 1970)
and p.height > (select avg(height) from players);
```

playerID	firstName	lastName	height	points
bryanko01	Kobe	Bryant	78	2832

Section 2. World (Make default database)

3. What are the countries with the highest number of different spoken languages? Show only the first 5 results.

(Hint: Group by Country name and sort properly and use LIMIT)

Solution:

```
select c.Name, count(*) as count from countrylanguage cl, Country c
where cl.CountryCode = c.Code
group by c.Name
order by count(*) DESC
Limit 5;
```

Name	count
Canada	12
Russian Federation	12
India	12
United States	12
China	12

4. Return for each continent, the population of its city having the lowest population.

Solution:

```
select ci.Name, min(ci.Population) as Population, c.Continent from
city ci, country c
where ci.CountryCode = c.Code
group by (c.Continent);
```

City	Population	Continent
Kabul	21484	Asia
Amsterdam	455	Europe
Willemstad	595	North America
Alger	1500	Africa
Tafuna	42	Oceania
Buenos Aires	1636	South America

